

Graphical Nature of Riemann Integral

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Overview

Riemann hypothesis for Integration was great, and it's phenomenal usage and logical purity is exquisite. It can be understood, simply and most intuitively as, **Area under the curve**: It

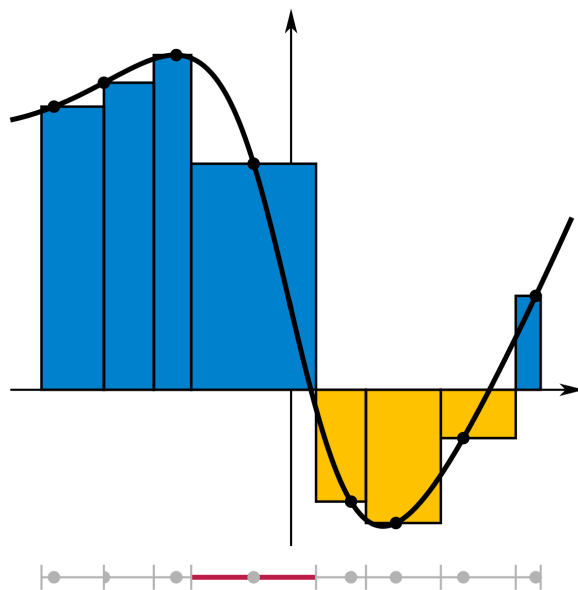


Figure 1: A Riemann-sum approximation to the integral of a function. The black curve represents the function, and the black dots mark the sample points used to determine the rectangle heights.

was intended to think it analytically, rather than just for computational basis. The formulation can be done as: Assuming a function $f(x)$ where x is in interval $[a, b]$, such that the interval is divided into k points where $P = \{x_0, x_1, x_2, x_3, \dots, x_k\}$. These points correspond to certain areas, where the **width is divided equally**. Accordingly, if we set k sufficiently big, we can get sufficiently close to the actual area under the curve. Assuming without loss of generality that $a = x_0 < x_1 < x_2 < x_3 < \dots < x_k = b$. Accordingly, we can formulate the area to be:

$$S(k) = \sum_{i=1}^k f(x_i)(x_{i+1} - x_i) \quad (1)$$

But, the analysis is not concluded here, after we put different such $f(x)$, like linear, exponential and else, this $S(k)$ changes.

Graphical Co-ordination of Integral Sum

Suppose a function $f(x) = 10 - x^2$, which is a downward parabolic curve with vertex $(0, 10)$. Firstly, let's get to a corollary of the "sum" expression in order to generalize and simplify necessary thing to accommodate change. Suppose the width of each rectangle is the same (equal), then we can assume that the total width i.e. $(b-a)$ is divided into say n equal lengths. From this we can say:

$$x_i = a + i\left(\frac{b-a}{n}\right), \quad y_i = f(x_i)$$

For instance we can assume, $a = 0$ and $b = x$, then the integral summation expression can be given as:

$$S(k) = \sum_{i=1}^k \left(\frac{b-a}{n}\right) f\left(i\left(\frac{b-a}{k}\right)\right) = \sum_{i=1}^k \left(\frac{b-a}{n}\right) f\left((i+1)\left(\frac{b-a}{n}\right)\right) \quad (2)$$

and the equality holds only for $k \rightarrow \infty$, else if k is known, defined or finite the expression in *RHS* is smaller than the *LHS*. To understand the nature of both of these expressions, the best choice would be "quadratic", as supposed before. Then:

$$S(k) = \sum_{i=1}^k \left(\frac{b-a}{n}\right) \left(10 - \left(i\left(\frac{b-a}{k}\right)\right)^2\right)$$

$$S'(k) = \sum_{i=1}^k \left(\frac{b-a}{n}\right) \left(10 - \left((i+1)\left(\frac{b-a}{k}\right)\right)^2\right)$$

From the above, expressions we can get to a graph:

This graph is applied for $n = 4$, and here the **red and blue graphs are $S'(k)$ and $S(k)$ respectively**. The significant part is the values of these two associative functions, for $x < 0$ and $x > 0$ the values of $S(k)$ is smaller and greater than $S'(k)$ respectively. Here, these two graphs come closer and closer as k increases. Hence, the result. This is our core result, that is, the one which is co-related to each other such that we get a relative idea of whole process. The exact graph can be found on: <https://www.desmos.com/calculator/tzmrhi6noo>, and the upcoming processes and engines are directed to GitHub.

Steps and values of "k"

If we analyze these graphs a little more actively, we can comment a regular periodic change in co-ordinates and overall function as k from the previous results increase. Let's take for instance the



Figure 2: Left: graphs of $S(k)$ (blue) and $S'(k)$ (red). Right: the accompanying animation.

same graph $S(k)$, which is:

$$S(k) = \sum_{i=1}^k \left(\frac{b-a}{n} \right) \left(10 - \left(i \left(\frac{b-a}{k} \right) \right)^2 \right)$$

The whole graphical movement is inscribed on GitHub Repo. Go to Change in Co-ordinaates of Curve to view the simulation or simply screen recording.

The most important thing in that is not the skeletal increase of graph but the way it vibrates (to and fro), it do that because the step magnitude is **0.1** and the value of k is any positive value upto first decimal place.